

Test for the mean. Population variance known

Data scientist salary

Background You are given the data from the lesson

Task 1 Test the null hypothesis at 10% significance

Dataset

\$ 117,313
\$ 104,002
\$ 113,038
\$ 101,936
\$ 84,560
\$ 113,136
\$ 80,740
\$ 100,536
\$ 105,052
\$ 87,201
\$ 91,986
\$ 94,868
\$ 90,745
\$ 102,848
\$ 85,927
\$ 112,276
\$ 108,637
\$ 96,818
\$ 92,307
\$ 114,564
\$ 109,714
\$ 108,833
\$ 115,295
\$ 89,279
\$ 81,720
\$ 89,344
\$ 114,426
\$ 90,410
\$ 95,118
\$ 113,382

Sample mean \$ 100,200
Population std \$ 15,000
Standard error \$ 2,739

Sample size 30

Z-score -4.67

H_0 (Glassdoor data) \$ 113,000
Two-sided test

The claim is that the average salary is \$113,000 and the statistical test showed that this null hypothesis should be rejected because it is not statistically significant i.e. it is strange for the observed values to fall at the extremes of the null distribution

2 Sided Test (When the null hypothesis contains an equal sign)

At 10% significance

Alpha 0.05
 $(1-\alpha) = (1-(0.1/2)) = 0.95$

Lookup 0.95 in z-table 1.65

Z score $|-4.67|$ is greater than 1.65
Therefore, reject the null hypothesis

